

The Kinetic–Event Synthesis (KES)

Mathematical Supplement & Structural Reference

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1. Core KES State

$$\mathfrak{S}_t = (X_t, \Omega_t, H_t, \Psi_t, \mathcal{M}_t, \mathcal{G}, \mathcal{F})$$

$$X_t = (\Phi_t, \mathbf{v}_t, S_t) \quad (\text{scalar, vector, entropy})$$

$$\Omega_t = \text{admissible future transitions}$$

$$H_t = \{e_0, e_1, \dots, e_t\} \quad (\text{irreversible event log})$$

$$\Psi_t = \text{preference / constraint field}$$

$$\mathcal{M}_t : \Omega_t \rightarrow \mathbb{R} \quad (\text{valuation measure})$$

$$\boxed{\Omega_t \longrightarrow H_{t+1}} \quad (\text{irreversible selection})$$

Reality = selection of possibility into history

2. RSVP Field Dynamics

Scalar plenum density:

$$\partial_t \Phi = D_\Phi \nabla^2 \Phi - \nabla \cdot (\Phi \mathbf{v}) + R_\Phi - \lambda_\Phi S$$

Vector flow (Navier–Stokes type):

$$\partial_t \mathbf{v} = -(\mathbf{v} \cdot \nabla) \mathbf{v} + \nu \nabla^2 \mathbf{v} - \nabla \Phi + \mathcal{T}(\mathbf{v})$$

Entropy production–dissipation:

$$\partial_t S = D_S \nabla^2 S + \sigma(\Phi, \mathbf{v}) - \gamma S$$

Coherence conditions:

$$\mathcal{U}_\omega \subset X_t, \quad \Phi \uparrow, \quad \|\mathbf{v}\|_{\text{aligned}} \uparrow, \quad S \downarrow$$

Admissibility field (Barbour extension):

$$\mathcal{A}_{\text{dm}}(X_t) = \{ \omega \in \Omega_t \mid \kappa(\omega, H_t) \geq \kappa_* \}$$

where κ is recursive admissibility and κ_* is the viability threshold. Admissibility replaces entropy as the fundamental cosmological selector.

3. Accessibility & Temporal Structure

Global model space (all kinematically possible states):

$$\mathfrak{M} = \{s_t \mid t \in \mathbb{R}\}$$

History-conditioned accessibility slice:

$$\mathcal{A}_t = \{s \in \mathfrak{M} \mid s \text{ reachable given } H_t\}$$

Temporal asymmetry (monotone shrinkage):

$$\mathcal{A}_{t+1}^{\text{consistent}} \subsetneq \mathcal{A}_t^{\text{counterfactual}}$$

The superscripts distinguish the *consistent* slice (states compatible with H_{t+1}) from the *counterfactual* envelope (states that were open before the selection at t).

Viability (non-collapse):

$$\Omega_t \neq \emptyset$$

Provenance-stable self invariant:

$$\Phi_{\text{self}} = \text{Inv}(H_t, \Omega_t, \mathcal{A}_t, \mathcal{P}_t)$$

Shape-space reduction (Barbour):

$$\mathfrak{M} / \sim_{\text{gauge}} \longrightarrow \mathcal{Q}_{\text{shape}}$$

Paths in $\mathcal{Q}_{\text{shape}}$ with maximal κ constitute the physical history.

4. Prompting as Field Projection

Constraint-valid structure space: $\omega \in \mathcal{S}_{\text{valid}}$

Projection functor:

$$\Pi : \mathcal{S}_{\text{valid}} \rightarrow \mathcal{L}$$

Rendering operator (stochastic):

$$\mathcal{R} : \mathcal{L} \rightsquigarrow \hat{\mathcal{L}}$$

Verification / grounding map:

$$\mathcal{V} : \hat{\mathcal{L}} \rightarrow \mathcal{S}_{\text{valid}} \cup \{\perp\}$$

Full pipeline:

$$\omega \xrightarrow{\Pi} p \xrightarrow{\mathcal{R}} \hat{\ell} \xrightarrow{\mathcal{V}} \omega'$$

Compression chain:

$$\text{Constraint} \rightarrow \text{Projection} \rightarrow \text{Rendering} \rightarrow \text{Verification}$$

5. Provenance & Identity

Provenance trace:

$$\mathcal{P}_t = \mathcal{P}(\omega_t), \quad \mathcal{P}_t \Rightarrow \omega_t$$

Identity as maximal integral curve:

$$\forall t, \quad \mathcal{P}_t \text{ reconstructs } \omega_t$$

Log sovereignty (state determined by history):

$$X_t = f(H_t)$$

Append-only event accumulation:

$$H_{t+1} = H_t \cup \{e_t\}$$

Irreversibility (no right inverse):

$$\Omega_t \rightarrow H_{t+1} \not\Rightarrow H_t$$

Upload incoherence: no isomorphism $\iota : X_t \xrightarrow{\sim} X'_t$ preserves H_t under substrate change; identity requires the integral curve, not merely the snapshot.

6. Expansion Operators

Expansion:

$$T : \Omega_t \rightarrow \Omega_{t+1}, \quad T(\Omega_t) \subseteq \Omega(H_{t+1})$$

Operator decomposition:

$$T = T_{\text{phys}} \circ T_{\text{comp}} \circ T_{\text{sym}} \circ T_{\text{sem}}$$

Spectral transform compatibility:

$$\mathcal{F}(\Omega_{t+1}) = \mathcal{T}(\mathcal{F}(\Omega_t))$$

Viability under composition:

$$\Omega_t \neq \emptyset \Rightarrow T(\Omega_t) \neq \emptyset$$

Intelligence = selection of viable expansion operators

7. Sheaf & Cohomology Structure

Let $\mathfrak{U} = \{U_i\}$ be an open cover of $\mathfrak{M}$. Local accessibility sheaf $\mathcal{A}$:

$$\mathcal{A}(U_i) = \mathcal{A}_t \cap U_i$$

Gluing condition (consistency of local histories):

$$\mathcal{A}(U_i)|_{U_i \cap U_j} = \mathcal{A}(U_j)|_{U_i \cap U_j}$$

Obstruction to global section:

$$[\delta\sigma] \in \check{H}^1(\mathfrak{U}, \mathcal{A})$$

Vanishing obstruction $\Leftrightarrow$ globally consistent history. Non-trivial $\check{H}^1$ encodes irreducible temporal contradictions or constraint-incoherent transitions.

Čech 2-cocycle (triple overlap):

$$\delta^2\sigma_{ijk} = \sigma_{jk} - \sigma_{ik} + \sigma_{ij} = 0$$

Holonomy of constraint transport around closed loop γ :

$$\text{Hol}_\gamma(\Psi) = \mathcal{P} \exp\left(\oint_\gamma A_\Psi d\ell\right) \in \text{Aut}(\Omega_t)$$

Non-trivial holonomy $\Rightarrow$ axial charge; cf. cosmic birefringence as global holonomy of the admissibility connection over $\mathcal{Q}_{\text{shape}}$.

8. Spherepop / KES Connection

Spherepop operators over event histories:

Operator	KES interpretation
Pop(e)	$H_t \rightarrow H_t \cup \{e\}$ (append event)
Bind(e, c)	$\omega \mapsto \omega'$ with constraint c
Collapse(ω)	$\Omega_t \rightarrow H_{t+1}$ (actuation)
Refuse(ω)	$\omega \notin \mathcal{A}_{\text{dm}}(X_t)$

Free traced monoidal structure:

$$(\mathbf{Sp}, \otimes, I, \text{Tr})$$

Trace encodes feedback loops; append-only H_t prevents trace from reversing events. The Spherepop category embeds into $\mathcal{C}$ via:

$$\iota : \mathbf{Sp} \hookrightarrow \mathcal{C}, \quad \iota(\text{Pop}) \mapsto [H_t \rightarrow H_{t+1}]$$

9. Economic Projection

Valuation measure:

$$\mathcal{M}_t : \Omega_t^{\text{agg}} \rightarrow \mathbb{R}$$

Lossy compression (distinct states collapse):

$$\mathcal{M}_t(\omega_1) = \mathcal{M}_t(\omega_2), \quad \omega_1 \neq \omega_2$$

Extraction regime (Goodhart divergence):

$$\Delta\mathcal{M}_t > 0, \quad \Delta\mathcal{R}_t < 0$$

Care functional:

$$\mathcal{C}_t : \mathfrak{S}_t \rightarrow \mathbb{R}, \quad \mathcal{M}_t \uparrow \Rightarrow \mathcal{C}_t \downarrow$$

Structural richness:

$$\mathcal{R}(\Omega) = \dim(\Omega) + \text{conn}(\Omega) + \text{coh}(\Omega)$$

Platform extraction reads as $\mathcal{M}_t$ maximisation with $\mathcal{R}(\Omega_{\text{worker}})$ as externality; austerity shrinks $\mathcal{A}_t^{\text{public}}$ while preserving $\mathcal{A}_t^{\text{capital}}$.

10. Markov Failure Case

Standard generator:

$$P(x_{n+1} | x_n), \quad H_t = \emptyset, \quad \mathcal{P}_t = \emptyset, \quad \mathcal{V} = \emptyset$$

Consequence:

$$\hat{\ell} \sim \text{prior manifold}$$

Surface novelty vs. structural distinction:

$$\delta_{\text{surface}}(\hat{\ell}_1, \hat{\ell}_2) > 0 \not\Rightarrow \delta_{\text{structural}}(\omega_1, \omega_2) > 0$$

Collapse regime: without H_t , outputs sample the attractor basin of the training distribution; constraint-valid generation requires $\mathcal{V} \neq \emptyset$.

11. Spectral Text Structure

Token embedding sequence:

$$x(n) \in \mathbb{R}^d, \quad n = 0, \dots, N-1$$

Discrete Fourier transform:

$$X(k) = \sum_{n=0}^{N-1} x(n) e^{-i2\pi kn/N}$$

Spectral signature:

$$\sigma_{\text{text}} = |X(k)|$$

Constraint-preserving transform:

$$f : \omega \rightarrow \omega', \quad \mathcal{C}(\omega) = 1 \Rightarrow \mathcal{C}(\omega') = 1$$

Low-frequency modes carry semantic coherence; high-frequency modes carry syntactic variation. Structural distinction lives in σ_{text} , not surface tokens.

12. Category Structure

Possibility category $\mathcal{C}$:

$$\text{Ob}(\mathcal{C}) = \{\Omega\}, \quad \text{Hom}(\Omega, \Omega') = \{\omega : \Omega \rightarrow \Omega'\}$$

History as composition:

$$H_t = \omega_t \circ \dots \circ \omega_1$$

Functors:

$$\Pi : \mathcal{S} \rightarrow \mathcal{L} \quad (\text{projection}), \quad \mathcal{M}_t : \mathcal{C} \rightarrow \mathbf{R} \quad (\text{valuation})$$

Natural transformation (power shift):

$$\eta : \mathcal{M}_t \Rightarrow \mathcal{M}'_t$$

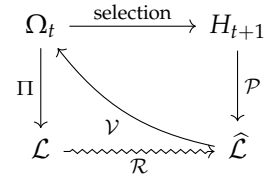
Alignment functor chain:

$$\mathbf{Val} \xrightarrow{F} \mathbf{Beh} \xrightarrow{G} \mathbf{Act}, \quad G \circ F \simeq \text{id}_{\mathbf{Val}} \quad (\text{alignment condition})$$

The weak equivalence $\simeq$ rather than strict equality acknowledges that behavioural implementation is at best naturally isomorphic to value specification; strict identity would require a fully transparent action–value correspondence unavailable under bounded rationality.

13. Master Diagram

The following square unifies the six primary KES relations—possibility, selection, provenance, projection, rendering, and verification—in a single commuting structure. The bent arrow $\mathcal{V}$ closes the loop from rendered output back into the constraint-valid possibility space, making the cycle explicit.



Reading the diagram: admissible possibilities Ω_t are selected into history H_{t+1} (top); simultaneously projected via Π into linguistic space $\mathcal{L}$ (left); stochastically rendered to $\hat{\mathcal{L}}$ (bottom); and verified by $\mathcal{V}$ back into the next possibility set (diagonal). Provenance $\mathcal{P}$ (right) records the selection event, coupling H_{t+1} to the rendered output and grounding the loop.

14. Fundamental Diagrams

Projection–rendering pipeline:

$$\omega \xrightarrow{\Pi} p \xrightarrow{\mathcal{R}} \hat{\ell} \xrightarrow{\mathcal{V}} \omega'$$

Temporal shrinkage:

$$\Omega_t \longrightarrow H_{t+1} \longrightarrow \mathcal{A}_{t+1}^{\text{consistent}} \subset \mathcal{A}_t^{\text{counterfactual}}$$

Full rendering cycle:

$$\mathcal{S}_{\text{valid}} \xrightarrow{\Pi} \mathcal{L} \xrightarrow{\mathcal{R}} \hat{\mathcal{L}} \xrightarrow{\mathcal{V}} \mathcal{S}_{\text{valid}} \cup \{\perp\}$$

Sheaf gluing pullback:

$$\begin{array}{ccc} \mathcal{A}(U_i \cup U_j) & \longrightarrow & \mathcal{A}(U_i) \\ \downarrow & & \downarrow \\ \mathcal{A}(U_j) & \longrightarrow & \mathcal{A}(U_i \cap U_j) \end{array}$$

15. Compression Statements

Time = constraint accumulation
Self = provenance-stable integral curve
Prompting = field-guided projection
Care = preservation of structural richness
Identity = maximal integral curve through H_t
Power = $\eta : \mathcal{M}_t \Rightarrow \mathcal{M}'_t$

Mind = local accessibility with provenance
Intelligence = coherence-preserving expansion
Economics = compressed possibility valuation
Reality = $\Omega_t \rightarrow H_{t+1}$
Cosmology = admissibility geometry on $\mathcal{Q}_{\text{shape}}$
Collapse = $\rho(D\mathcal{G}_{\mathfrak{E}^*}) > 1$

16. Recursive Constraint Geometry

Constraint propagation over admissible trajectories:

$$\mathcal{C}_{t+1}(\omega) = \mathcal{C}_t(\omega) + \Delta_{\text{hist}}(\omega, H_t) - \Delta_{\text{entropy}}(\omega)$$

Recursive admissibility flow:

$$\kappa_{t+1}(\omega) = \kappa_t(\omega) + \int_{\gamma_\omega} \Psi_t \cdot d\ell - S_t(\omega)$$

Constraint curvature tensor:

$$\mathfrak{R}_{ij} = \partial_i \Psi_j - \partial_j \Psi_i + [\Psi_i, \Psi_j]$$

Flat constraint regions satisfy $\mathfrak{R}_{ij} = 0$; nonzero curvature yields path-ordering dependence:

$$\mathfrak{R}_{ij} \neq 0 \Rightarrow \omega_a \circ \omega_b \neq \omega_b \circ \omega_a$$

Corollary 0.1. *Historical ordering is dynamically significant precisely when constraint curvature is non-vanishing.*

Corollary 0.2. *Regions of high recursive curvature correspond to strongly irreversible civilizational or cognitive transitions.*

17. Provenance Entropy & Information Flow

Provenance entropy:

$$S_{\mathcal{P}}(H_t) = - \sum_i p_i \log p_i$$

where p_i is the recoverability weight of provenance branch i .

Recoverability functional:

$$\mathcal{Q}_{\text{rec}} = \frac{\dim(\text{Recoverable}(H_t))}{\dim(H_t)}$$

Epistemic fragmentation condition:

$$\partial_t \mathcal{Q}_{\text{rec}} < 0$$

Information transport equation:

$$\partial_t I + \nabla \cdot (I \mathbf{v}) = D_I \nabla^2 I - \lambda_I S$$

Constraint-preserving channels satisfy:

$$\frac{d}{dt} \mathcal{C}(\omega_t) \geq 0$$

Corollary 0.3. *Compression systems maximising transmission while minimising provenance necessarily increase semantic ambiguity.*

Corollary 0.4. *A civilisation with declining provenance recoverability will increasingly substitute symbolic authority for reconstructible derivation.*

18. Narrative Type Theory

Let $\mathbb{T}$ be the category of admissible narrative types. Objects $\tau \in \text{Ob}(\mathbb{T})$ are stable narrative invariants; morphisms $f : \tau_i \rightarrow \tau_j$ are admissible identity transitions.

Narrative typing judgment:

$$H_t \vdash \tau$$

Consistency condition:

$$H_t \vdash \tau_i \wedge \tau_i \not\sim \tau_j \Rightarrow H_t \not\vdash \tau_j$$

Narrative bifurcation operator:

$$\mathcal{B} : \tau \mapsto \{\tau_1, \tau_2, \dots\}$$

Self-coherence invariant:

$$\chi_{\text{self}} = \int_{H_t} \text{Compat}(\tau_t, \tau_{t+1}) dt$$

Corollary 0.5. *Psychological fragmentation corresponds to decreasing path-compatibility within the narrative type manifold.*

Corollary 0.6. *Stable identity is not static persistence but coherent type evolution through irreversible history.*

19. Spectral Constraint Topology

Decompose $\Omega_t = \bigcup_i \Omega_i$ into local admissibility domains.
Spectral Laplacian:

$$\Delta_\Omega = D - A$$

where D is the degree operator and A the accessibility adjacency.

Eigenstructure:

$$\Delta_\Omega \phi_n = \lambda_n \phi_n$$

Low-frequency modes ($\lambda_n \approx 0$) encode globally coherent semantic structures; high-frequency modes ($\lambda_n \gg 0$) encode localised fluctuations or instability.

Constraint resonance condition:

$$\langle \phi_i, \Psi_t \rangle \gg 0$$

Phase-locking:

$$\phi_i(t + \Delta t) \approx e^{i\theta} \phi_i(t)$$

Corollary 0.7. *Long-term civilisations stabilise around low-frequency admissibility eigenmodes.*

Corollary 0.8. *Memetic instability corresponds to excitation of high-frequency spectral components exceeding damping capacity.*

20. The KES Fixed-Point Theorem

Recursive evolution operator:

$$\mathfrak{S}_{t+1} = \mathcal{G}(\mathfrak{S}_t), \quad \mathfrak{S}^* = \mathcal{G}(\mathfrak{S}^*)$$

Local stability via spectral radius ρ :

Condition	Regime
$\rho(D\mathcal{G}_{\mathfrak{S}^*}) < 1$	Stable expansion
$\rho \approx 1$	Metastable
$\rho > 1$	Collapse

Theorem 0.9 (Recursive Coherence Theorem). *A KES trajectory remains structurally viable iff recursive admissibility, provenance recoverability, and expansion coherence remain jointly bounded.*

Sketch. If admissibility collapses, the trajectory exits $\mathcal{A}_{\text{dm}}(X_t)$. If provenance collapses, reconstruction of ω_t from $\mathcal{P}_t$ fails. If expansion coherence collapses, Ω_t fragments into disconnected components. Joint boundedness is therefore necessary for persistent structural identity. $\square$

Corollary 0.10. *Intelligence is not maximal optimisation but bounded coherent continuation through recursively constrained possibility space.*

Corollary 0.11. *The terminal failure mode of civilisations is topological fragmentation of their admissibility manifold.*